Complete Swift XRT Spectral Analysis Manual

A Step-by-Step Guide for XSPEC

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Introduction

This manual provides a comprehensive guide to analyzing Swift XRT data using XSPEC. It covers the standard workflow: loading data, defining physical models (GRB Afterglows), fitting, error analysis, flux calculation, and plotting.

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1 Part 1: Setup and Data Loading

Before performing any scientific analysis, you must load the data files and inspect the quality of the spectrum.

1.1 1. Navigate and Start

Open your terminal in the directory containing your processed files (e.g., `interval0`) and start the program:

```
xspec
```

1.2 2. Load the Data

Load your spectrum file. If your data has > 400 counts, use the grouped file. If it is faint, use the raw file.

```
XSPEC12> data interval0pc_grp.pi
```

Note: XSPEC usually automatically detects and loads the Background (.back), Response Matrix (.rmf), and Ancillary Response (.arf) files. If not, load them manually using ***back****,* ***resp****, and* ***arf****.*

1.3 3. Set the Statistic

Choose the statistical method based on your data quality:

- **Chi-Squared (Standard):** Use for grouped data (> 20 counts/bin).

  ```
  XSPEC12> statistic chi
  ```

- **C-Stat (Low Counts):** Use for faint sources.

  ```
  XSPEC12> statistic cstat
  ```

1.4 4. Set the Energy Range (Crucial)

The Swift XRT is calibrated reliably only between 0.3 keV and 10.0 keV. You must ignore data outside this range.

```
XSPEC12> ignore **-0.3 10.0-**
```

For the mature missions in the HEASARC archives, this information already has been encoded in the spectral files using a "QUALITY" column according to the standard described in OGIP Memo OGIP/92-007. Simply issue the command:

```
 XSPEC12> ignore bad
```

1.5 5. Set Plotting Device

```
XSPEC12> cpd /xs   % For Linux/macOS
XSPEC12> cpd /gw   % For Windows
```

2 Part 2: Defining the Model

We use the standard GRB Afterglow model: **Galactic Absorption × Intrinsic Absorption × Powerlaw Source**.

2.1 1. Define the Structure

```
XSPEC12> model phabs*zphabs*powerlaw
```

- `phabs`: Photoelectric absorption from the Milky Way (foreground).
- `zphabs`: Photoelectric absorption from the host galaxy (redshifted).
- `powerlaw`: Synchrotron emission from the GRB source.

2.2 2. Entering Parameters

XSPEC will prompt you for values. Use the table below as a guide:

Component	Parameter	Value to Enter	Explanation
`phabs`	nH	`0.0416, -1`	Galactic N_H (from HEASARC nH Tool). Divide by 10^{22}. The `-1` freezes it.
`zphabs`	nH	`0.2`	Intrinsic N_H guess. Leave positive to fit.
`zphabs`	Redshift	`1.14, -1`	Enter z if known. If unknown, use 0. The `-1` freezes it.
`powerlaw`	PhoIndex	`2.0`	Standard GRB slope guess.
`powerlaw`	norm	`(Enter)`	Press Enter to use the default start value.

3 Part 3: Fitting

3.1 1. Renormalize

Adjusts the model brightness to roughly match the data before mathematical minimization begins.

```
XSPEC12> renorm
```

3.2 2. Run the Fit

Minimizes the statistic (Chi-Squared or C-Stat) to find the best parameter values.

```
XSPEC12> fit
```

Check: Look for "Reduced chi-squared" in the output. A value between 0.8 and 1.2 indicates a good fit.

3.3 3. Calculate Errors

Always calculate the 90% confidence region for your variable parameters.

```
XSPEC12> error 2 4
```

4 Part 4: Modifying Parameters

4.1 Freezing (Locking a parameter)

To stop a parameter from changing during a fit:

```
XSPEC12> freeze 2
% OR manually set the step size to negative:
XSPEC12> newpar 2 0.5 -1
```

4.2 Thawing (Unlocking a parameter)

To allow a parameter to vary during a fit:

```
XSPEC12> thaw 2
```

5 Part 5: Flux Calculation

5.1 A. Observed Flux (Instrumental)

This is the flux actually hitting the telescope (dimmed by gas).

```
% Syntax: flux [min_E] [max_E] err [simulations] [confidence]
XSPEC12> flux 0.3 10.0 err 100 90
```

5.2 B. Unabsorbed Flux (Physical)

To determine the intrinsic luminosity, you must calculate the flux as if no gas were present. The most robust method is using the `cflux` convolution model.

Step 1: Edit the model Wrap the powerlaw component in `cflux`.

```
XSPEC12> editmod phabs*zphabs*cflux(powerlaw)
```

Step 2: Set parameters

- `Emin`: 0.3
- `Emax`: 10.0
- `lg10Flux`: -11 (Guess). Ensure this parameter is **free**.

Step 3: Refit Ensure your absorption parameters (N_H) are set correctly (Galactic frozen, Intrinsic free), then fit.

```
XSPEC12> fit
```

The `lg10Flux` parameter now represents the log of the unabsorbed flux.

6 Part 6: Plotting

6.1 1. Standard Fit Plot

Displays the data against the model and the residuals (errors).

```
XSPEC12> setplot energy
XSPEC12> plot data delchi
```

6.2 2. Physical Spectrum (E^2F_E)

Displays the "Unfolded Spectrum" to show the energy distribution.

```
XSPEC12> setplot energy
XSPEC12> plot eeufspec
```

Note : Up to six options can be given to the plot command at a time to generate plot with vertically stacked windows. Example :

```
XSPEC12> plot data delchi eeufspec
```

6.3 3. Rescaling Axes

If the plot limits are incorrect, use the interactive plotter (`iplot`).

```
XSPEC12> iplot
PLT> rescale y 1e-4 1e-1    % Use small positive numbers for log scales
PLT> exit
```

6.4 4. Saving the Plot

To export a publication-quality PostScript file:

```
XSPEC12> iplot
PLT> hardcopy spectrum_plot.ps/ps
PLT> exit
```

6.5 Binning Data (Visual Grouping)

To group data points in the plot window for better visibility (without affecting the statistical fit), use the `setplot rebin` command.

```
% Syntax: setplot rebin <sigma> <max_bins>
% Example: Combine bins to achieve 3-sigma significance, max 10 bins
   combined
XSPEC12> setplot rebin 3 10
```

6.6 Changing Y-Axis to Log Scale

To change the Y-axis from linear to logarithmic (essential for viewing faint tails in spectra), use `setplot ylog`.

```
% Switch Y-axis to Logarithmic scale
XSPEC12> setplot ylog

% Switch Y-axis back to Linear scale
XSPEC12> setplot ylinear

% Refresh the plot to see changes
XSPEC12> plot data ratio
```

6.6.1 Alternative: Using Interactive Plotter

If the standard command does not refresh the window immediately, force it using `iplot`:

```
XSPEC12> iplot
PLT> log y      % Force Log Scale
PLT> plot       % Refresh
PLT> exit
```